

3 Pin Definitions

For explanations of pin names and function names, as well as configurations of peripheral pins, please refer to [ESP32-S3 Series Datasheet](#).

Table 3-1. Pin Definitions

Name	No.	Type ^a	Function
GND	1, 2, 42, 43, 46-65	P	GND
3V3	3	P	Power supply
IO0	4	I/O/T	RTC_GPIO0, GPIO0
IO1	5	I/O/T	RTC_GPIO1, GPIO1, TOUCH1, ADC1_CH0
IO2	6	I/O/T	RTC_GPIO2, GPIO2, TOUCH2, ADC1_CH1
IO3	7	I/O/T	RTC_GPIO3, GPIO3, TOUCH3, ADC1_CH2
IO4	8	I/O/T	RTC_GPIO4, GPIO4, TOUCH4, ADC1_CH3
IO5	9	I/O/T	RTC_GPIO5, GPIO5, TOUCH5, ADC1_CH4
IO6	10	I/O/T	RTC_GPIO6, GPIO6, TOUCH6, ADC1_CH5
IO7	11	I/O/T	RTC_GPIO7, GPIO7, TOUCH7, ADC1_CH6
IO8	12	I/O/T	RTC_GPIO8, GPIO8, TOUCH8, ADC1_CH7, SUBSPICS1
IO9	13	I/O/T	RTC_GPIO9, GPIO9, TOUCH9, ADC1_CH8, FSPIHD, SUBSPIHD
IO10	14	I/O/T	RTC_GPIO10, GPIO10, TOUCH10, ADC1_CH9, FSPICSO, FSPIIO4, SUBSPICSO
IO11	15	I/O/T	RTC_GPIO11, GPIO11, TOUCH11, ADC2_CH0, FSPID, FSPIIO5, SUBSPID
IO12	16	I/O/T	RTC_GPIO12, GPIO12, TOUCH12, ADC2_CH1, FSPICLK, FSPIIO6, SUBSPICLK
IO13	17	I/O/T	RTC_GPIO13, GPIO13, TOUCH13, ADC2_CH2, FSPIQ, FSPIIO7, SUBSPIQ
IO14	18	I/O/T	RTC_GPIO14, GPIO14, TOUCH14, ADC2_CH3, FSPIWP, FSPIDQS, SUBSPIWP
IO15	19	I/O/T	RTC_GPIO15, GPIO15, UORTS, ADC2_CH4, XTAL_32K_P
IO16	20	I/O/T	RTC_GPIO16, GPIO16, UOCTS, ADC2_CH5, XTAL_32K_N
IO17	21	I/O/T	RTC_GPIO17, GPIO17, U1TXD, ADC2_CH6
IO18	22	I/O/T	RTC_GPIO18, GPIO18, U1RXD, ADC2_CH7, CLK_OUT3
IO19	23	I/O/T	RTC_GPIO19, GPIO19, U1RTS, ADC2_CH8, CLK_OUT2, USB_D-
IO20	24	I/O/T	RTC_GPIO20, GPIO20, U1CTS, ADC2_CH9, CLK_OUT1, USB_D+
IO21	25	I/O/T	RTC_GPIO21, GPIO21
IO26 ^b	26	I/O/T	SPICS1, GPIO26
IO47 ^b	27	I/O/T	SPICLK_P_DIFF, GPIO47, SUBSPICLK_P_DIFF
IO33	28	I/O/T	SPIIO4, GPIO33, FSPIHD, SUBSPIHD
IO34	29	I/O/T	SPIIO5, GPIO34, FSPICSO, SUBSPICSO
IO48	30	I/O/T	SPICLK_N_DIFF, GPIO48, SUBSPICLK_N_DIFF
IO35	31	I/O/T	SPIIO6, GPIO35, FSPID, SUBSPID
IO36	32	I/O/T	SPIIO7, GPIO36, FSPICLK, SUBSPICLK
IO37	33	I/O/T	SPIDQS, GPIO37, FSPIQ, SUBSPIQ
IO38	34	I/O/T	GPIO38, FSPIWP, SUBSPIWP
IO39 ^a	35	I/O/T	MTCK, GPIO39, CLK_OUT3, SUBSPICS1
IO40	36	I/O/T	MTDO, GPIO40, CLK_OUT2
IO41	37	I/O/T	MTDI, GPIO41, CLK_OUT1

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Name	No.	Type ^a	Function
IO42	38	I/O/T	MTMS , GPIO42
TXD0	39	I/O/T	U0TXD , GPIO43, CLK_OUT1 } → Don't use CLK further at boot
RXD0	40	I/O/T	U0RXD , GPIO44, CLK_OUT2
IO45	41	I/O/T	GPIO45 EN } → Don't use
IO46	44	I/O/T	GPIO46
EN	45	I	High: on, enables the chip. Low: off, the chip powers off. Note: Do not leave the EN pin floating. } → En / Reset P2

^a P: power supply; I: input; O: output; T: high impedance. Pin functions in bold font are the default pin functions.

For pin 28 ~ 29, 31 ~ 33, the default function is decided by eFuse bit.

^b For modules with part numbers ending with -N4R2, IO26 connects to the embedded PSRAM and is not available for other uses.